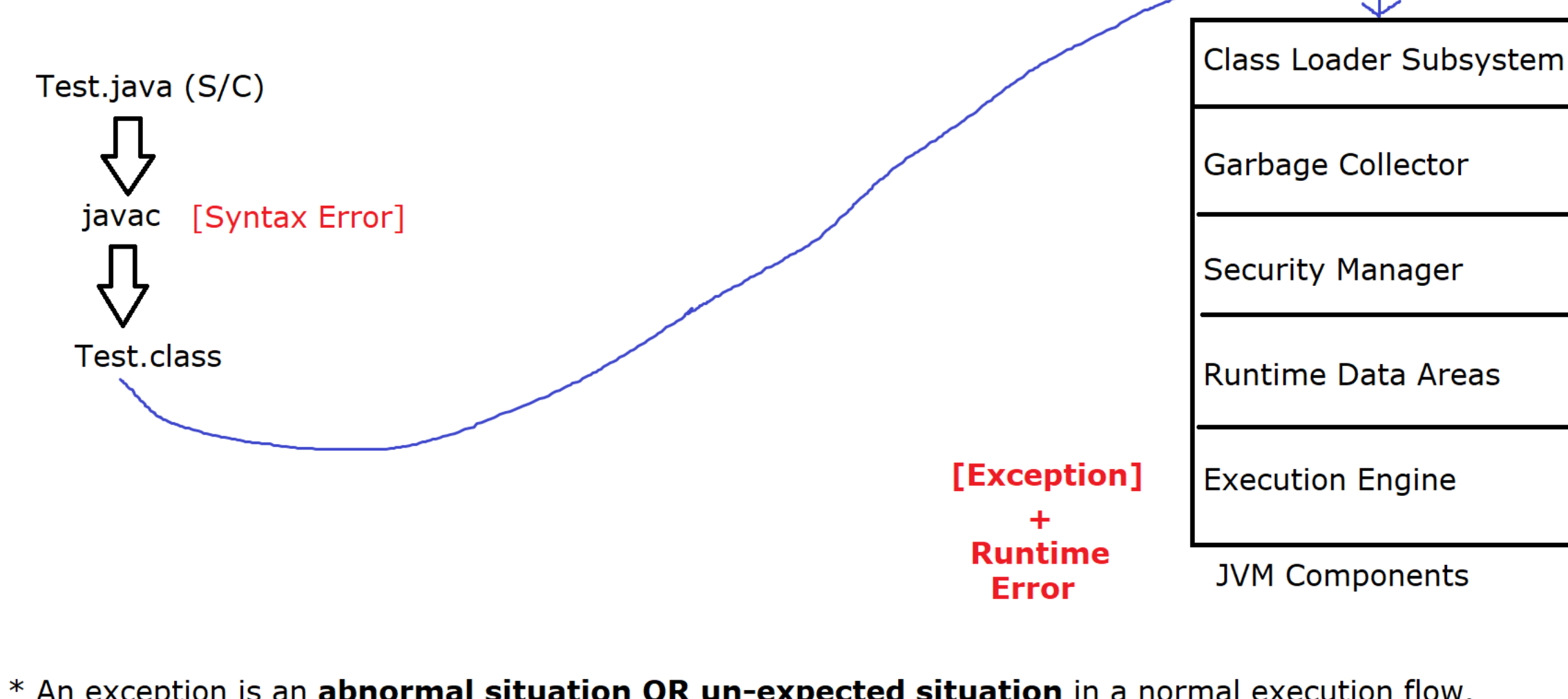


Exception Handling :

* An exception is a **Runtime error**.

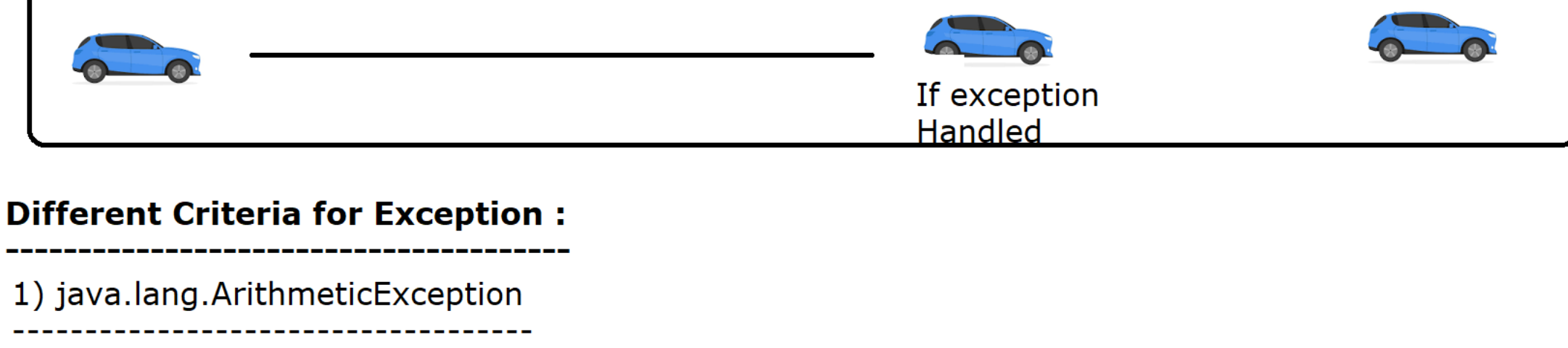


* An exception is an **abnormal situation OR un-expected situation** in a normal execution flow.

* Due to an exception the execution of the program will be terminated, if not handled properly.

* An exception is encountered due to the following two reasons :

- Wrong input given by the end user
- Due to dependency [One part of the program is dependent to another part]



Different Criteria for Exception :

1) java.lang.ArithmeticException

Whenever we divide a number by 0 (int value) then we will get an exception i.e java.lang.ArithmeticException

Example :
int x = 100;
int y = 0;
int z = x / y; //java.lang.ArithmeticException

2) java.lang.ArrayIndexOutOfBoundsException

Whenever we try to access the index of the array where index is not available then we will get java.lang.ArrayIndexOutOfBoundsException

Example :
int [] arr = {10,20,30};
System.out.println(arr[3]);

3) java.lang.NegativeArraySizeException :

The size of an array **must be positive integer** (only int value is acceptable), if we pass -ve size then we will get an exception java.lang.NegativeArraySizeException

Example:
int [] arr = new int[-10];

4) java.lang.NumberFormatException

If we try to convert any String value into corresponding primitive type OR Wrapper type and if the String data is not available in **numeric format** then we will get an exception java.lang.NumberFormatException.

Example :
String str = "Java";
int val = Integer.parseInt(str);
IO.println(val);

String s1 = "NIT";
Integer x = Integer.valueOf(s1);
IO.println(x);

5) java.lang.NullPointerException :

If a reference variable is pointing to null and if we try to invoke any non static method or non static field by using reference variable then we will get java.lang.NullPointerException

Case 1 :

String str = null;
IO.println(str.length());

Case 2 :

Scanner sc = new Scanner(System.in);
IO.print("Enter your Name :");
String name = sc.nextLine(); //null
IO.println("Name in upper case "+name.toUpperCase()); //NULL

Case 3 :

String name = IO.readLine("Enter your Name :"); //null
IO.println("Name in upper case "+name.toUpperCase()); //NULL

6) java.util.InputMismatchException :

If we try to read the data by using Scanner class methods and if the data is not in a proper format/input then we will get exception java.util.InputMismatchException

Example : nextInt() method will only read numeric int value

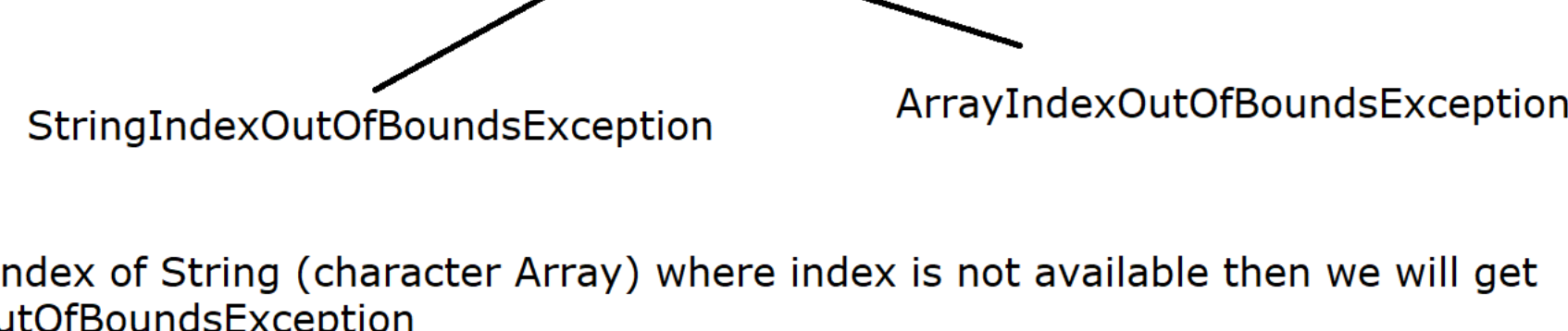
Case 1 :

Scanner sc = new Scanner(System.in);
IO.print("Enter your Roll number :");
int roll = sc.nextInt();
IO.println("Your roll number is :"+roll);

Case 2 :

Scanner sc = new Scanner(System.in);
System.out.println("Enter your Salary :");
double salary = sc.nextDouble(); //Eleven thousand [Exception]
System.out.println("Your Salary is :"+salary);

7) java.lang.StringIndexOutOfBoundsException :



If we try to access the index of String (character Array) where index is not available then we will get java.lang.StringIndexOutOfBoundsException

Example :

String str = "Hello";
IO.println(str.charAt(9));

Note : ArrayIndexOutOfBoundsException class StringIndexOutOfBoundsException class both are sub class of java.lang.IndexOutOfBoundsException.

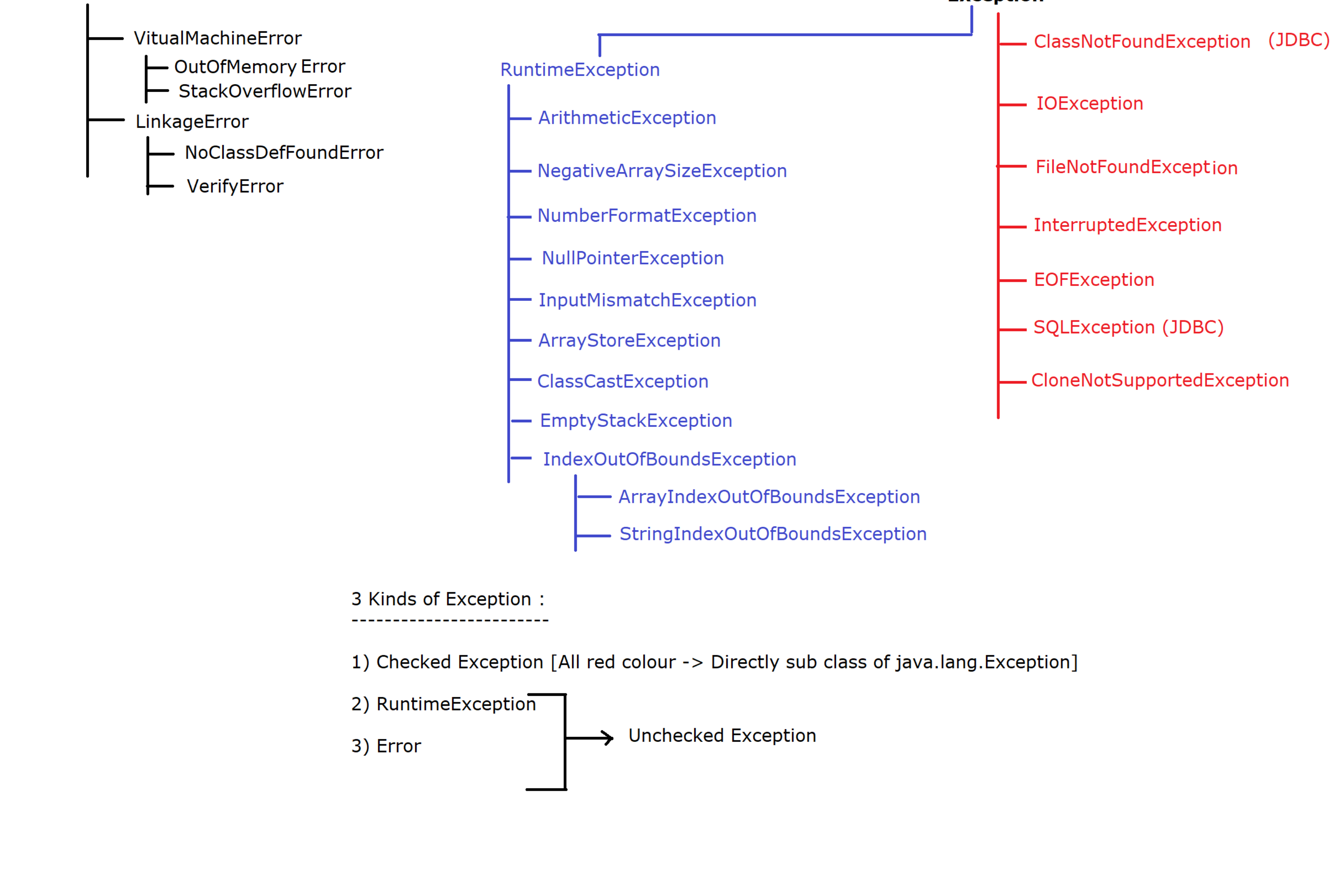
8) java.lang.ArrayStoreException :

If we try to insert some illegal data in the array at runtime then we will get a runtime exception i.e java.lang.ArrayStoreException

Example :
Object [] arr = new String[3];
arr[0] = "Java";
arr[1] = "HTML";
arr[2] = 90; //java.lang.ArrayStoreException

for(Object x : arr)
{
 IO.println(x);
}

Exception Hierarchy :



3 Kinds of Exception :

1) Checked Exception [All red colour -> Directly sub class of java.lang.Exception]

2) RuntimeException

3) Error

Unchecked Exception